Computer Vision
Lecture 8: LiDAR Concept

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History of LiDAR



In the 1930s, EH Synge conceived of LiDAR.

Columbia University team led by Charles Townes build the **first maser** (microwave amplification by stimulated emission of radiation)



The Hughes Mark II Colidar is the **first commercial LiDAR.**

George D. Hickman flies the first **bathymetric** LiDAR.

York University (Canada) physicist Allan Carswell and his wife Helen found **Optech.**

Milton Huffaker founds Coherent Technologies to commercialize LiDAR **wind-**detection systems

University of Texas at Austin team led by James Gibeaut create the first LiDAR-based digital elevation model of an **archaeological** site, at Copan, Honduras.

1930 1939 1953 1960 1962 1963 1968 1969 1974 1978 1984 1998 2000 2004

Ellis A. Johnson, a Carnegie Institution Scientist, uses a searchlight to capture signals up to 40 km.

In 1960, TH Maiman conceived the **first laser.**



SRI build the SRI Mark I LiDAR for **atmospheric** studies.



Apollo Lunar Ranging Experiment (LURE) lands on the moon with **Apollo 11.**



Johannes Riegl founds Riegl.

RIEGL

Cyra Technologies introduces the Cyrax 2400, the first tripod-mounted, commercial **3D scanner.**

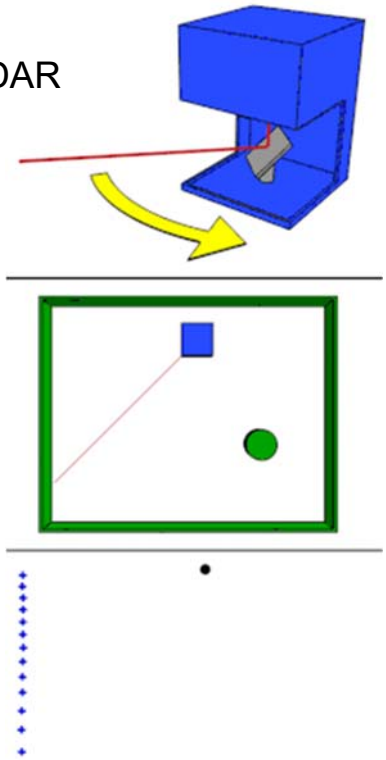
CYRA

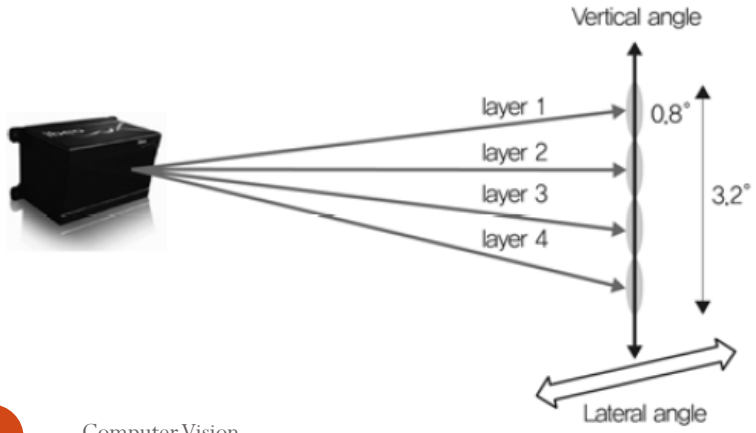
DARPA Grand Challenge in the Mojave Desert ends with no finishers. SICK line scanners are a popular sensor among teams.

LiDAR_for_Automotive_and_Industrial_Applications_Yole_2020_Report

LiDAR

- Light Detection And Ranging
- or Laser Detection And Ranging
- 905–1550 nm lasers are most common for LiDAR
- 1D/2D/3D, 3D: Multi-layer Technology
- Eye-safety : 1550nm
- Distance / Reflection intensity

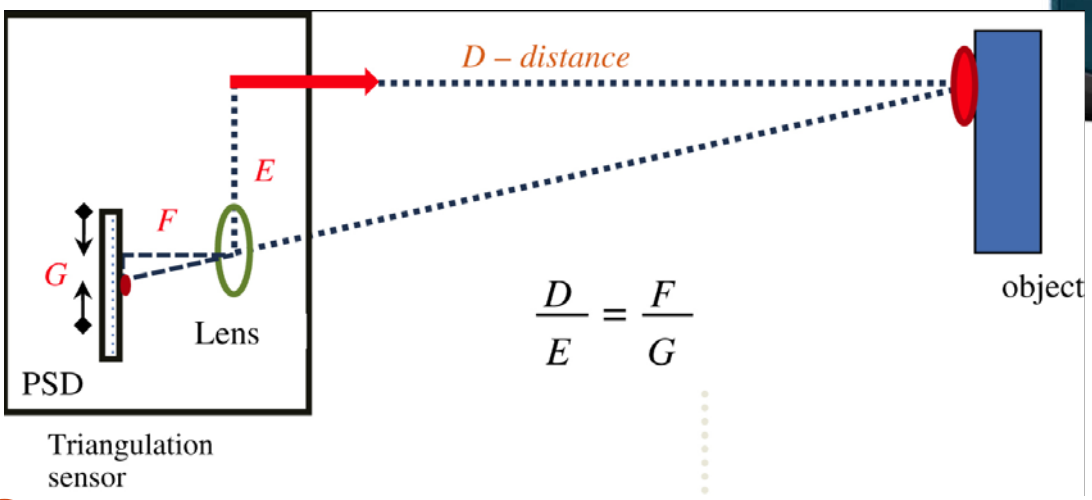




Triangulation 1D Lidar

- PSD: Position-sensitive detector

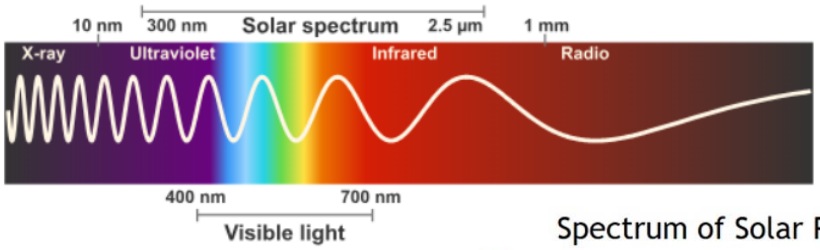




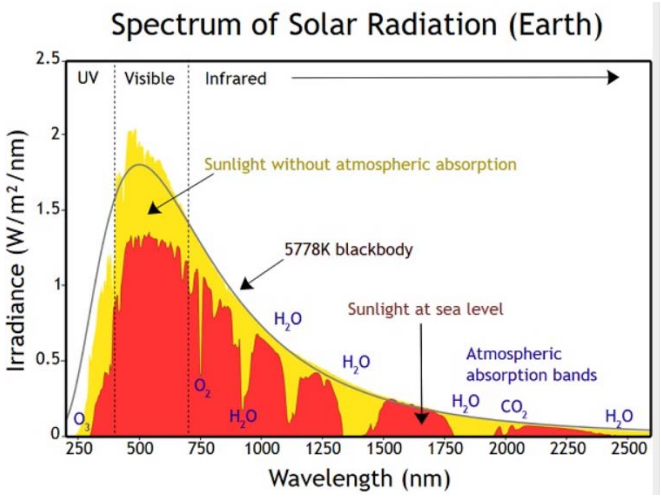
$$\frac{D}{E} = \frac{F}{G}$$

Sun light

About 99% of the solar power reaching Earth's surface has a wavelength between 300 nm and 2.5 μm.



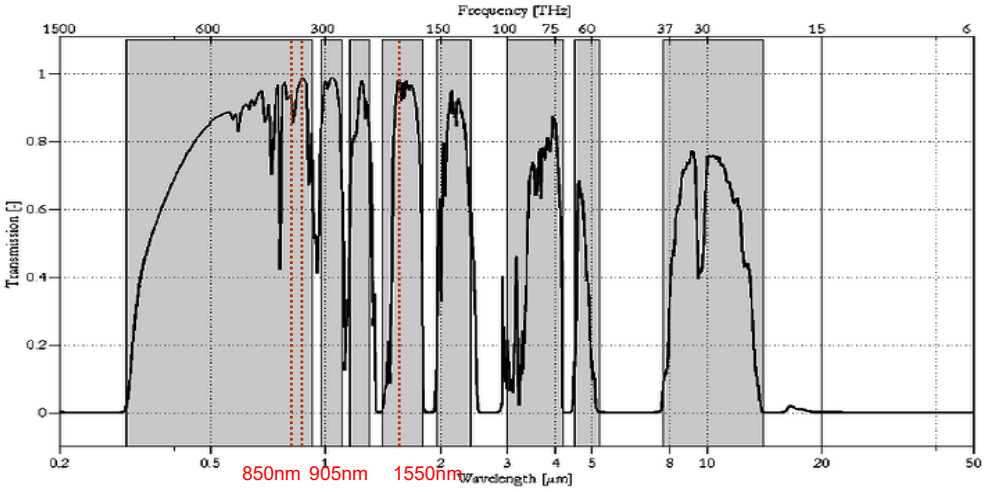
https://brilliant.org/practice/spectral-properties-sunlight/



https://www.solarquotes.com.au/blog/uv-solar-panels/

Optical attenuation in air

- Attenuation peaks due to absorption by water, oxygen, etc.
- Wavelength used in Lidar: 850nm, 905nm, 1550nm
- Wavelength choices by atmospheric attenuation and laser/photodetector technology maturity, availability and cost
- 850nm Lidar is being rapidly phased out



https://www.researchgate.net/profile/Otakar-Wilfert/publication/225005720/figure/fig1/AS:302674777001984@1449174698195/Atmospheric-transmittance-based-on-absorption-analysis-using-LOWTRAN-13-A-zenith-path.png

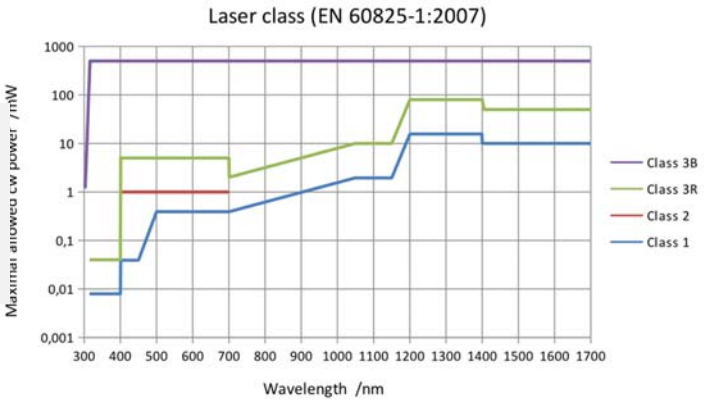
Lidar laser wavelengths

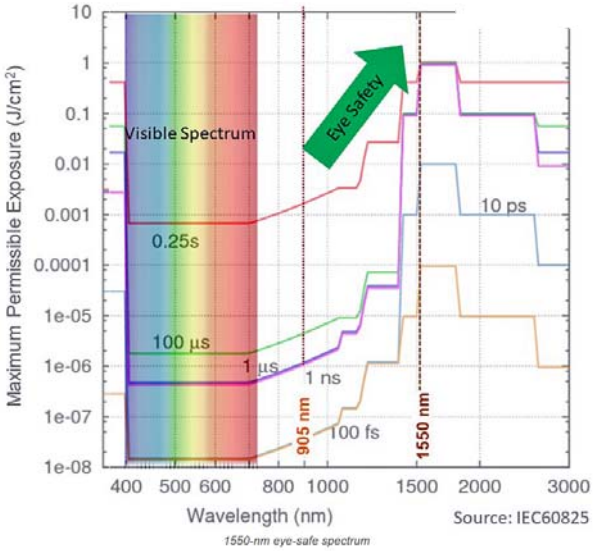
- Two main wavelengths used in commercial Lidars:
- 905 nm
  - Very common in direct detection(a.k.a. TOF) Lidar
  - Cheaper components (lasers, photodetectors, etc.)
  - Lower absorption by water (helps with bad weather performance)
  - Higher solar radiation at 905nm (more interference)
  - Lower photodetector responsivity and photon detection efficiency
  - Lower maximum permissible exposure(MPE) limit for eye safety (less output power)
- 1550 nm
  - Mostly used in coherent Lidar
  - Higher MPE limit enables higher power
  - Less interference from solar radiation
  - Higher photodetector responsivity and photon detection efficiency
  - More expensive components
  - Higher absorption by water

https://www.researchgate.net/profile/Otakar-Wilfert/publication/225005720/figure/fig1/AS:302674777001984@1449174698195/Atmospheric-transmittance-based-on-absorption-analysis-using-LOWTRAN-13-A-zenith-path.png

Eye Safety

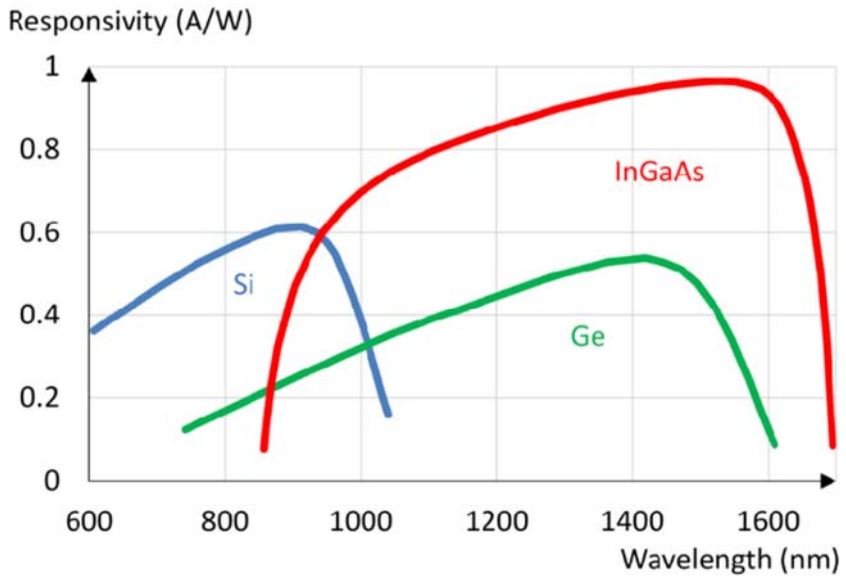
- Class 1/ 1M / 2/ 2M/ 3R/ 3B/ 4
- 1,550nm: eye-safety 면에서 우수하나 가격이 비쌈





https://www.fierceelectronics.com/components/1550-nm-photonics-promise-eye-safe-cost-effective-autonomous-machines

Photodetector responsivity along optical wavelength



13. Typical responsivity vs wavelength for silicon, InGaAs, and germanium.

- 게르마늄 (**Ge**)
- 인듐(In), 갈륨(Ga), 비소(As)

https://www.researchgate.net/figure/Typical-responsivity-vs-wavelength-for-silicon-InGaAs-and-germanium_fig2_347881095

905 nm laser

- Wavelength:

$$\lambda = 905\ \text{nm} = 905 \times 10^{-9}\,\text{m}$$

- Speed of light:

$$c \approx 3 \times 10^{8}\,\text{m/sec}$$

- Frequency :

$$f = \frac{c}{\lambda} = \frac{3 \times 10^{8}\text{m/sec}}{905 \times 10^{-9}\text{m}} = 3.314 \times 10^{14}\text{Hz} = 331\text{Terahertz}$$

- Period:

$$T = \frac{1}{f} = 2.38 \times 10^{-15} sec = 2.38\ \text{femto(펨토)}$$

Light comparison[10]			
Name	Wavelength	Frequency (Hz)	Photon energy (eV)
Gamma ray	less than 0.01 nm	more than 30 EHz	more than 124 keV
X-ray	0.01 nm – 10 nm	30 EHz – 30 PHz	124 keV – 124 eV
Ultraviolet	10 nm – 400 nm	30 PHz – 790 THz	124 eV – 3.3 eV
Visible	400 nm–700 nm	790 THz – 430 THz	3.3 eV – 1.7 eV
Infrared	700 nm – 1 mm	430 THz – 300 GHz	1.7 eV – 1.24 meV
Microwave	1 mm – 1 meter	300 GHz – 300 MHz	1.24 meV – 1.24 μeV
Radio	1 meter – 100,000 km	300 MHz – 3 Hz	1.24 μeV – 12.4 feV

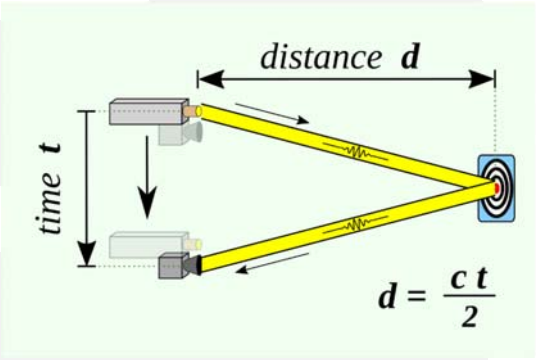
ToF(Time of Flight)

- Measurement principle used for imaging using LiDAR
- Depth measured by the round-trip delay of light waves that travel to the target
- Distance :

$$d = \frac{c \times \Delta t}{2},\ c \approx 3 \times 10^8 \text{m/sec}\ \text{ is the speed of light}$$

  - 200m round-trip :

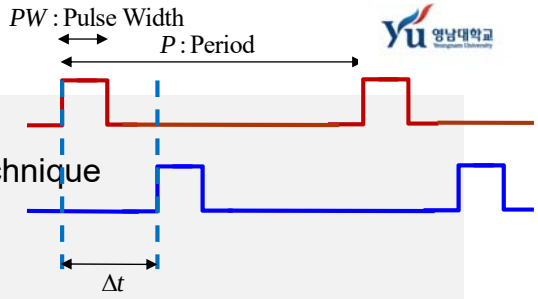
$$\Delta t = \frac{2d}{c} = \frac{400}{3\times10^8} = 1.33\mu\text{sec}$$



Distance Measuring methods

- Pulse
  - Single pulse
  - Multi-pulses
- AMCW(amplitude modulation of continuous wave)
- FMCW(frequency-modulated of continuous wave)

Pulse approach



- Single pulse : simplest modulation
  - Direct time-of-flight measurement technique
  - Duty cycle: $D = \frac{PW}{P}$
  - Pulse width: 수~수십 nsec

Assuming maximum distance to be measured: d_{max} =200 m
round trip time: $\Delta t_{max} = 1.33\mu\text{sec}$

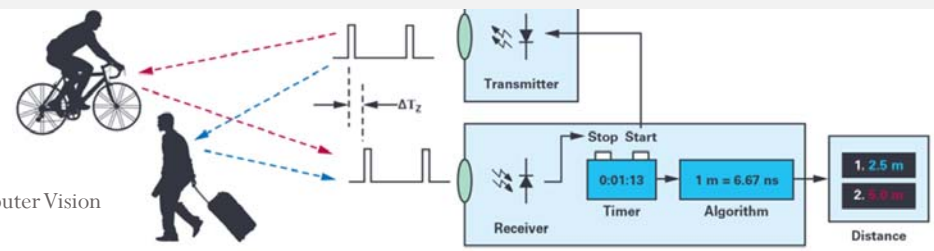
  - Pulse period:

$$P > \Delta t_{max}$$

  - Attainable resolution in range depends on time counting available
    - 만약 0.1 nsec (905nm 33,149 주기)오차는 약 3 cm 거리 (1.5cm , round trip)오차 발생

$$1\text{ sec} : 3\times10^8\text{ m} = 0.1\times10^{-9}\text{ sec} : x$$
$$x = 3\times10^8 \times 0.1\times10^{-9} = 0.03m = 3cm$$

- Multi Pulse : Single pulse 보다 정확하나 다소 복잡



Pulse approach

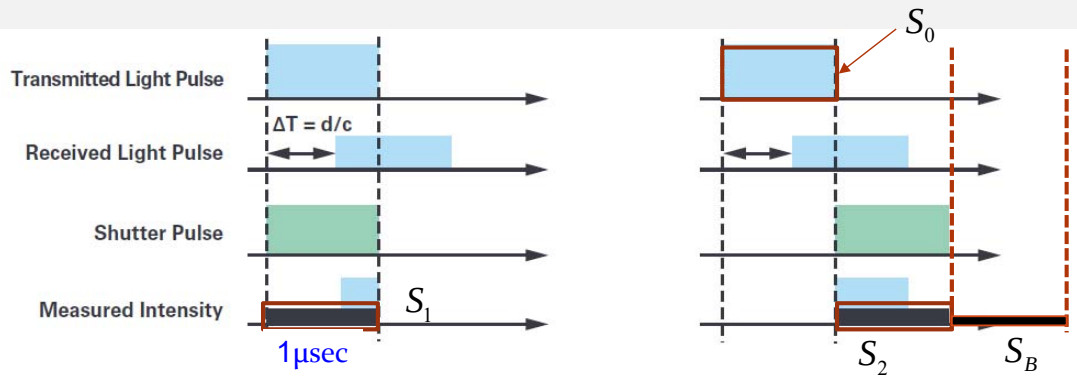
- EX: A single pulse TOF measurement method using CMOS
  - Assume the electric shutter time of CMOS and the pulse width be the same (for example, 1000 nsec= 1μsec)
  - 3 shutter time is required

round trip time: $\Delta t_{max} = 1\mu\sec$

$\Rightarrow d = \frac{c \times \Delta t}{2} = 150 meter$

$$\Delta t = \frac{S_2 - S_B}{S_1 + S_2 - 2S_B} \times T_{shutter} \Rightarrow d = \frac{\Delta t \times c}{2} = \frac{S_2 - S_B}{S_1 + S_2 - 2S_B} \times T_{shutter} \times \frac{c}{2}$$

$$reflction = \frac{S_1 + S_2}{S_0}$$

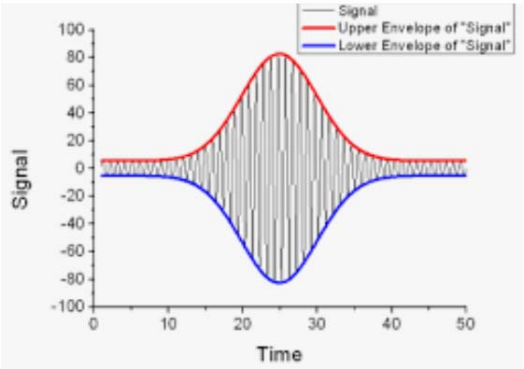


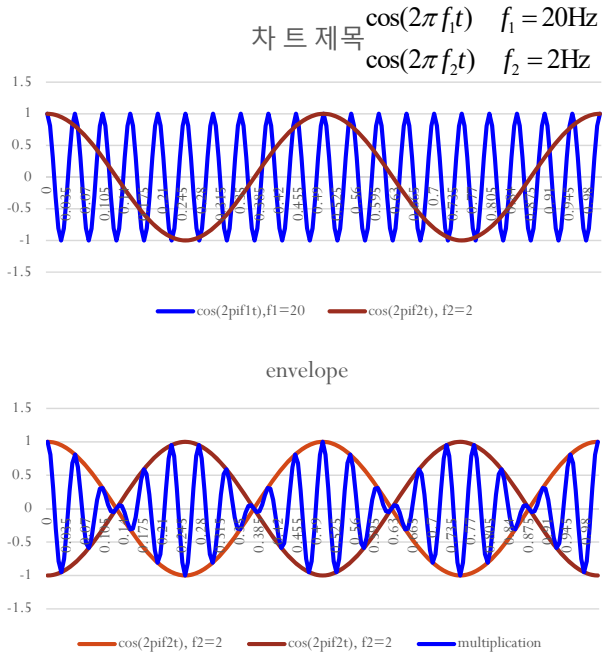
수학: 정현파의 곱

- Multiplication of two cosine function

$$\cos(2\pi f_1 t) \times \cos(2\pi f_2 t)$$

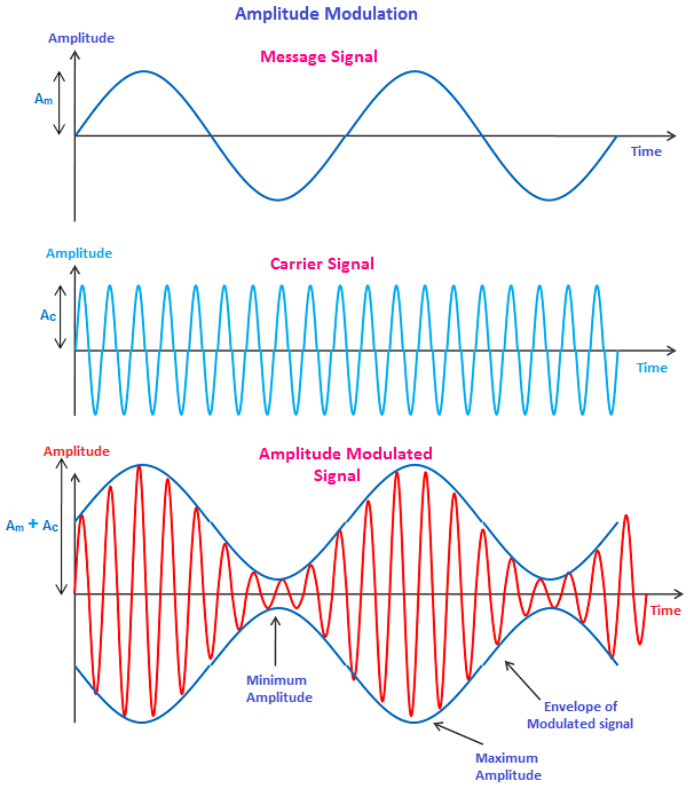
- Envelope





통신: AM (amplitude modulation)

- Amplitude modulation 이용



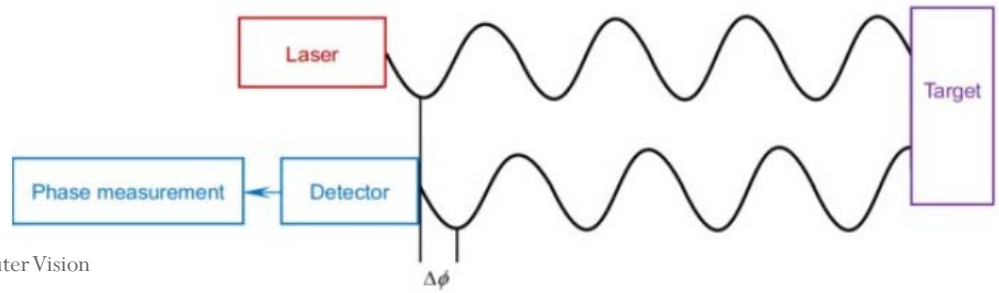
3. 국내 AM 방송 (1978년 이후)
   - ㅇ 주파수대역 : 535 ~1,605kHz 까지의 주파수대역
   - ㅇ 채널간격 : 9 kHz
   - ㅇ 채널 수 : 120 개

AMCW(amplitude modulation of continuous wave)

- Amplitude modulation 이용
- Phase measurement
  - Transmitted: $\cos(2\pi f_M t)$
  - Received: $\cos(2\pi f_M t - \triangle\varphi) = \cos(2\pi f_M (t - \triangle t))$
  - Phase meter

$$\triangle\varphi = 2\pi f_M \triangle t \;\Rightarrow\; d = \frac{c \times \triangle t}{2} = \frac{c\,\triangle\varphi}{4\pi f_M}$$

  - Methods: mixers and low-pass filters, cross-correlation, etc.



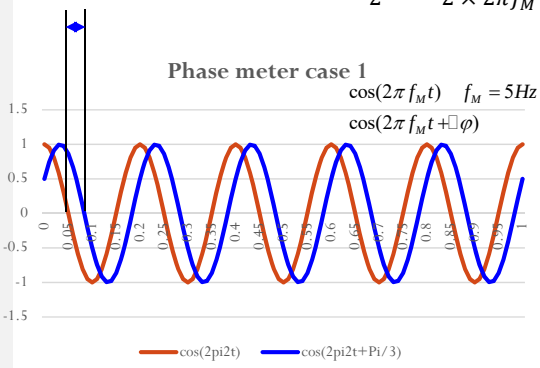
AMCW(amplitude modulation of continuous wave)

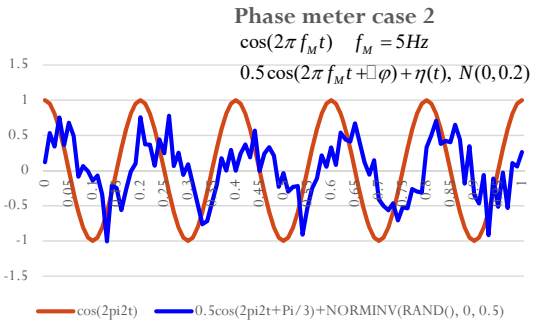
- Resolution is increased by increasing the modulation frequency f_M
- However, larger f_M brings on shorter unambiguous range measurements
  $\triangle\varphi, 2\pi+\triangle\varphi, 4\pi+\triangle\varphi, \cdots$
- Typical modulation frequencies are generally in the few tenths MHz range f_M

  if $d_{\max}$=200 m,

  round trip time: $\Delta t_{\max} = 1.33\mu\sec$

$$\triangle\varphi = 2\pi f_M \triangle t \;\Rightarrow\; d = \frac{c\times\triangle t}{2} = \frac{c\times\triangle\varphi}{2\times 2\pi f_M}$$





수학: Beat frequency

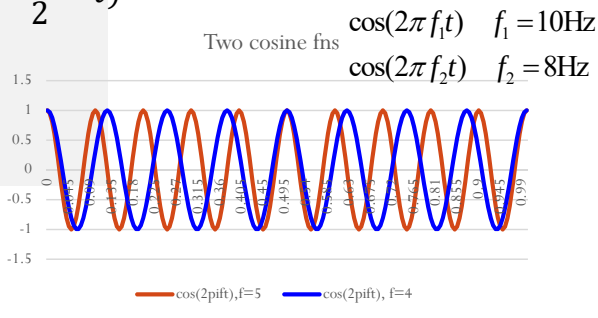
- Addition(interference) of two cosine functions with slightly different frequencies

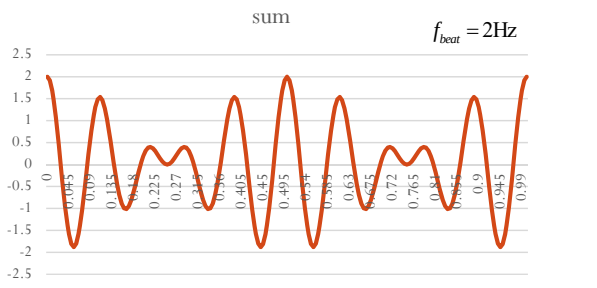
$$\cos(2\pi f_1 t) + \cos(2\pi f_2 t) = 2\cos(2\pi\frac{f_1 - f_2}{2}t)\cos(2\pi\frac{f_1 + f_2}{2}t)$$

- Beat frequency

$$f_{beat} = |f_1 - f_2|$$

$$2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \sin\alpha + \sin\beta$$
$$2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) = \sin\alpha - \sin\beta$$
$$2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \cos\alpha + \cos\beta$$
$$-2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) = \cos\alpha - \cos\beta$$

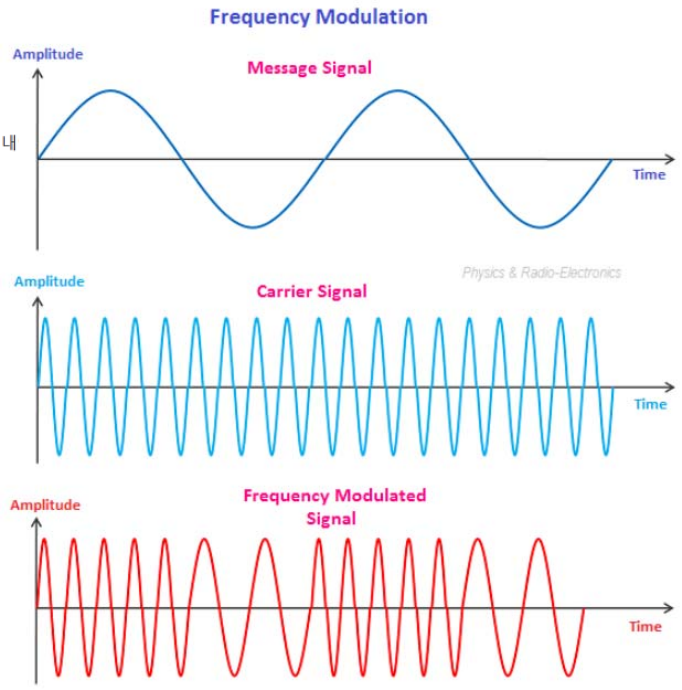




통신: FM (Frequency modulation)

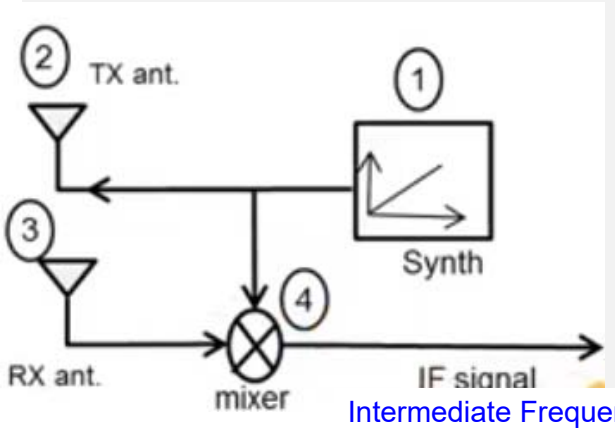
- Frequency modulation 이용

3. 국내 FM 방송
  - ㅇ 주파수대역 : **88 ~ 108MHz(VHF)**
    - VHF TV 채널의 저대역과 고대역 사이인 88 ~ 108MHz에서 20MHz 범위내 모두 100개 채널로 배분
  - ㅇ 채널간격 : 200 kHz
  - ㅇ 최대 주파수 이탈 편차 : +75 kHz



FMCW(Frequency modulation of continuous wave)

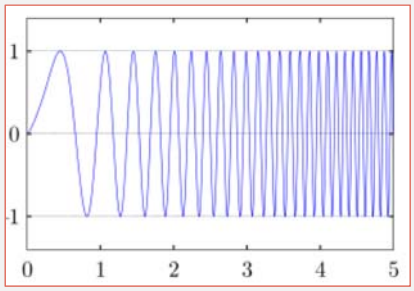
- 기본구성



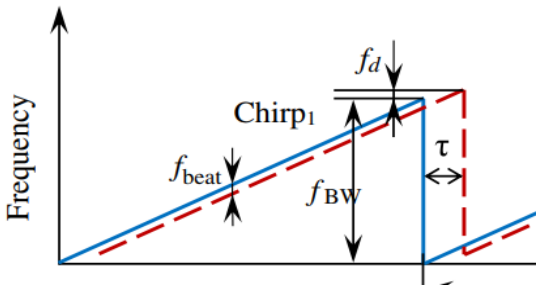
Intermediate Frequency

Transmitter :

chirp : "A-t plot" Amplitude vs time



chirp "f-t plot" Frequency vs time



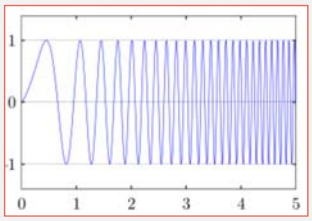
FMCW(Frequency modulation of continuous wave)

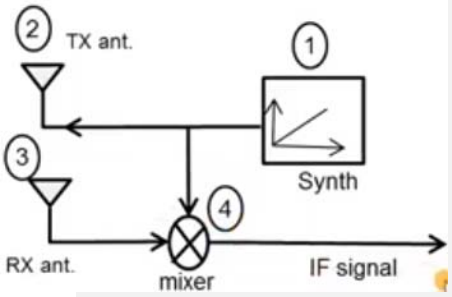
- Frequency modulation 이용

Transmitter :

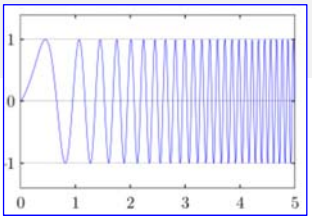
chirp

"A-t plot" Amplitude vs time



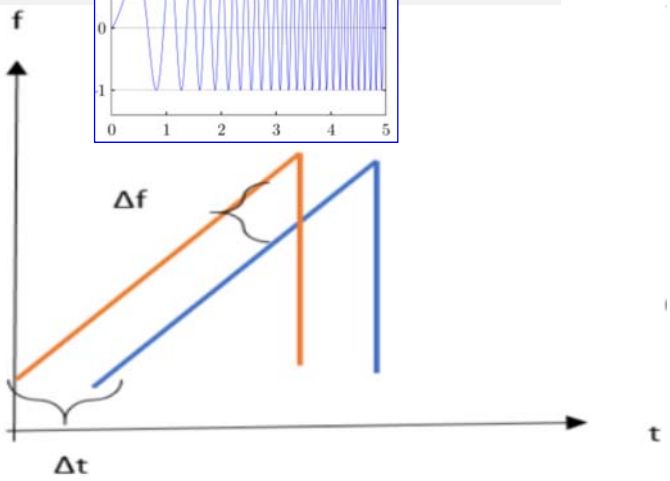


Receiver :



chirp

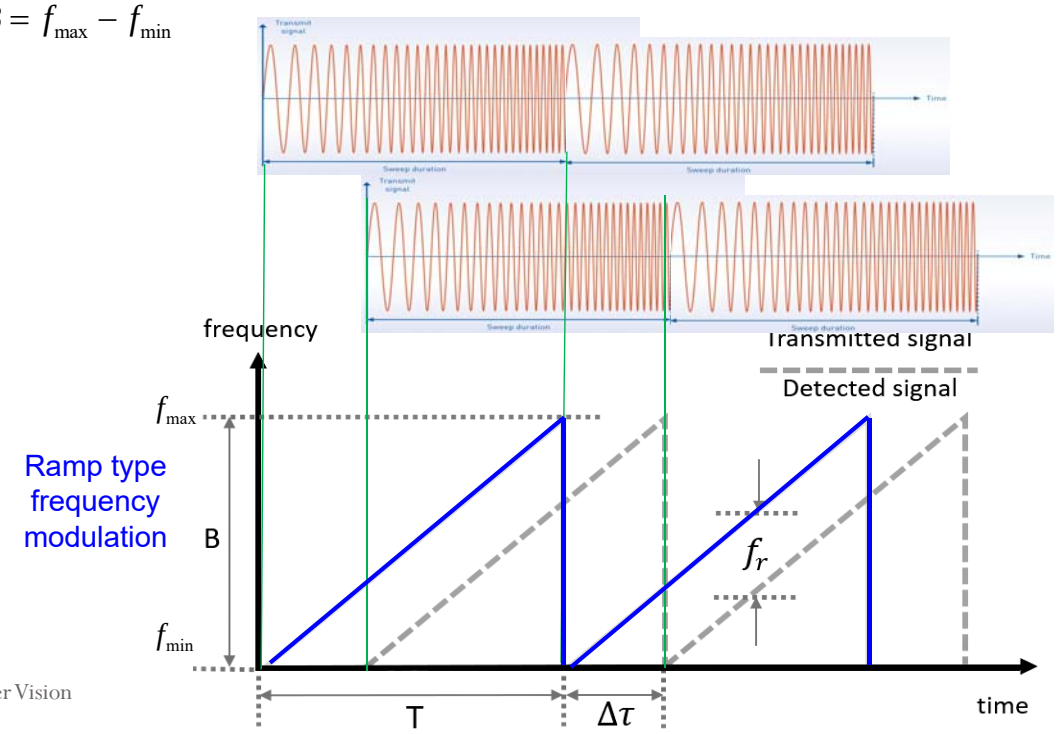
"f-t plot" Freauency vs time



FMCW(Frequency modulation of continuous wave)

$$B:T = f_r:\Delta\tau \;\Rightarrow\; \Delta\tau = \frac{f_r T}{B} \;\Rightarrow\; R = \frac{c\Delta\tau}{2} = \frac{cf_r T}{2B}$$

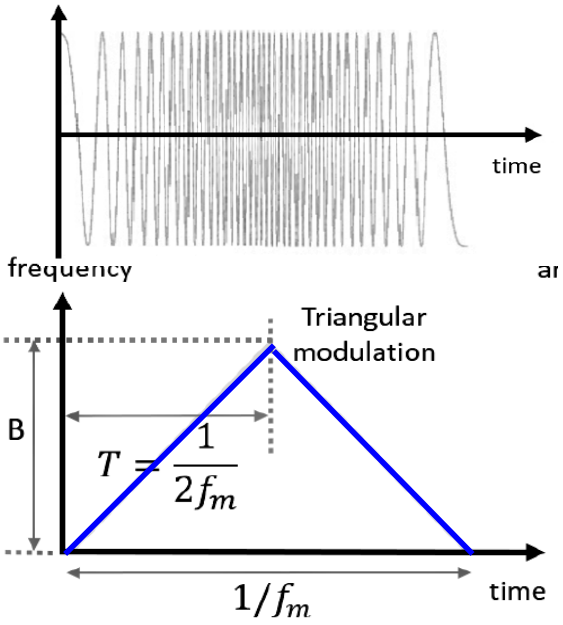
where $B = f_{\max} - f_{\min}$



FMCW(Frequency modulation of continuous wave)

$$R = \frac{cf_rT}{2B} \overset{T = \frac{1}{2f_m}}{\Rightarrow} R = \frac{cf_r}{4f_mB} \Rightarrow f_r = \frac{4Rf_mB}{c}$$

Triangular type frequency modulation



FMCW(Frequency modulation of continuous wave)

Doppler Effect

- The relationship between observed frequency and emitted frequency is given by

c is the propagation speed of light

v is the speed of target

$$f_{re} = \left(\frac{c \pm v}{c \mp v}\right) f_{tr}$$

$$\Rightarrow f_d = f_{re} - f_{tr} = \left(\frac{c \pm v}{c \mp v} - 1\right) f_{tr} = \left(\frac{c \pm v - (c \mp v)}{c \mp v}\right) f_{tr} = \left(\frac{\pm 2v}{c \mp v}\right) f_{tr}$$

$$\Rightarrow f_d \approx \frac{\pm 2v f_{tr}}{c}, \quad \text{if } c \mp v \to c \quad \Rightarrow v \approx \frac{cf_d}{\pm 2f_{tr}} = \frac{\lambda_{tr} f_d}{\pm 2}$$

- Note the frequency will decrease if either source or target is moving away from the other.

FMCW(Frequency modulation of continuous wave)

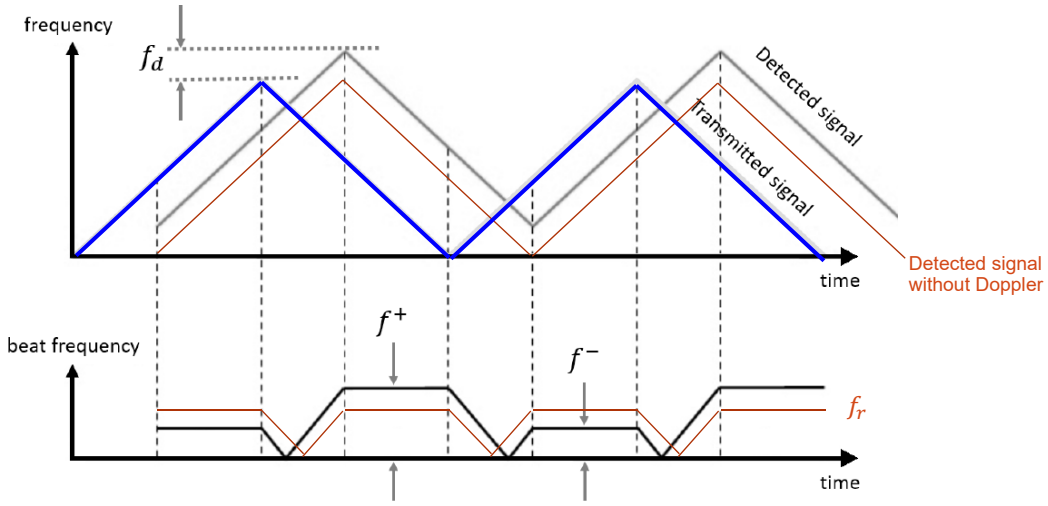
f_d : Doppler frequency

$$f^+ = f_r + f_d \\ f^- = f_r - f_d \Rightarrow f_r = \frac{f^+ + f^-}{2} \Rightarrow R = \frac{cf_r}{4f_m B} = \frac{c}{8f_m B}(f^+ + f^-)$$

$$v \approx \frac{\lambda_{tr}}{2} f_d = \frac{\lambda_{tr}}{4}(f^+ - f^-)$$: Relative velocity

속도 200K/h => T=10μsec 동안 0.56 mm 이동

$$\lambda_{tr} = \frac{c}{f_{\text{tr}}}, f_{\text{tr}} \in B$$



FMCW(Frequency modulation of continuous wave)

$$f^+ = f_r + f_d \\ f^- = f_r - f_d \Rightarrow f_r = \frac{f^+ + f^-}{2} \Rightarrow R = \frac{cf_r}{4f_m B} = \frac{c}{8f_m B}(f^+ + f^-)$$

$$v \approx \frac{\lambda_{tr}}{2} f_d = \frac{\lambda_{tr}}{4}(f^+ - f^-) \qquad \lambda_{tr} = \frac{c}{f_{\text{tr}}}, f_{\text{tr}} \in B$$

- EX: Radar parameters: center frequency $f_{\text{tr}} = 24\ GHz$ ($\lambda_{tr} = \frac{c}{f_{\text{tr}}} = \frac{3\times10^8}{24\times10\text{^}9} = 0.0125\ m$), bandwidth $B = 200\ MHz$, modulation frequency $f_m = 250\ Hz$. Observed data: beat frequency on rising interval $f^+ = 120\ KHz$, beat frequency on decreasing interval $f^- = 80\ KHz$.
- What is the distance?

$$R = \frac{c}{8f_m B}(f^+ + f^-) = \frac{3\times10^8}{8\times250\times200\times10^6}(120\times10^3 + 80\times10^3) = 150\ meters$$

- What is the target's velocity?

$$v \approx \frac{\lambda_{tr}}{4}(f^+ - f^-) = \frac{0.0125}{4}(120\times10^3 - 80\times10^3) = \frac{125m}{sec} = 450Km/hour$$

FMCW(Frequency modulation of continuous wave)

- Positives
  - Resolution is related to total bandwidth (B) : Theoretically millimeter or even submillimeter distances can be measured
  - Ability to sense simultaneously the speed value with its direction together with range
- Negatives
  - A perfect linear or triangular optical frequency sweep cannot be realized
  - Coherent detection scheme needed

Comparison of Distance Measuring methods (ToF)

	Pulsed	AMCW	FMCW
Parameter measured	Intensity of emitted and received pulse	Phase of modulated amplitude	Relative beat of modulated frequency, and Doppler shift
Measurement	Direct	Indirect	Indirect
Detection	Incoherent	Incoherent	Coherent
Use	Indoor/Outdoor	Only indoor	Indoor/Outdoor
Main advantage	Simplicity of setup; long ambiguity range	Established commercially	Simultaneous speed and range measures
Main limitation	Low SNR of returned pulse	Short ambiguity distance	Coherence length/Stability in operating conditions (e.g., thermal)
Depth resolution (typ)	1 cm	1 cm	0.1 cm

Thank You
Next : LiDAR Imaging

